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The Noisy Work of Uncertainty Visualization

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data visualization, statistical graphics, data science

Abstract

Better representation of the uncertainty in a data visualization is a focus of recent research activity. A problem with the current literature is that there is a lack of clarity about the definition of uncertainty and what it means to represent it in a plot. This confusion has led to conflicting results in the literature, especially in experiments that assess the effectiveness of different uncertainty representations. In this review, we summarize the current literature, provide workable definitions, and illustrate these definitions with examples. In doing so, we ask what it really takes to achieve transparency in statistical graphics. We hope that it will be useful for guiding new graphics methodology and experimental research.

A Mathematical Framework and Software Implementation for Uncertainty Visualisation

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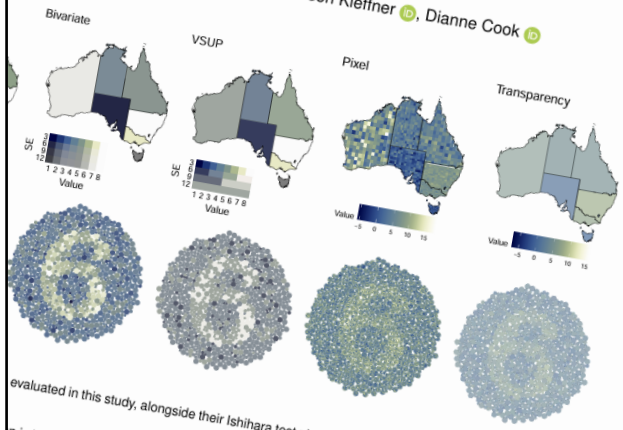
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Abstract

Random variables are the bread and butter of statistics, and visualisations are of the most versatile tools in the field, so it is a wonder why there is no agreed n methodology for visualising random variables. This gap is particularly evident xploratory data analysis (EDA). We address this gap by designing a mathematical ework for visualisation, which argues that we should consider visualisations to ontinuous functions. In the case of random variable inputs, this means our isations should obey the continuous mapping theorem (CMT). By breaking the function down into its components, we are able to identify which parts of the g are ill-defined for random variable inputs. We then redefine the function y that guarantees both the flexibility required for EDA and the statistical ty of CMT. This formalisation represents a complete integration of uncertainty grammar of graphics, which we show by implementing the theory in the ge, `ggdibbler`. The ggdibbler software is a `ggplot2` extension that allows replace the data of any plotting function with a random variable, with the that the visualisation will have the same convergence properties as its g data.

Colour Blinded by the Noise

Harriet Mason, Rachel Rogers, Alison Kleffner, Dianne Cook



evaluated in this study, alongside their Ishihara test plate equivalent. Which is the better representation

n is important for data transparency, especially for map visualisations where data is often aggregated. a, studies evaluating uncertainty visualisation lack consensus and produce conflicting results. This pproach for uncertainty visualisation that attempts to assess uncertainty as noise, rather than signal. ising uncertainty: standard choropleth maps, value/variance bivariate maps, value-suppressing ling, and pixelated sampling maps. Built on principles of implicit testing, we put an 'uncertainty ara colourblind test to create a novel test that is able to evaluate uncertainty as noise. We compare hesis tests at various levels of group separation. By building our experimental design on top of te the plot components that facilitate successful signal suppression and establish foundational y visualisation.

ization, graphical testing, statistical graphics

ed in data visualisation. ns we draw about our nces for our analysis. -making" framework n [40]. A surprising n making" umbrella, scoveries [11], [33], tracting or compar- formation (such as

missing values) [47]. On the surface, these goals appear harmonious; *however*, an underlying conflict prevents a visualisation from achieving all of them simultaneously.

According to Mason et al. [36], the current decision-making framework conflates the two dichotomous roles of uncertainty in analysis: signal and noise. Signal is the information that we hope to extract from a plot, while noise is the interference that prevents us from extracting said signal. These dual roles are a by-product of the fact that uncertainty visualisations can be described as plots that have replaced their deterministic data input with a random matrix, where every individual cell is now a distribution [25], [36]. Uncertainty acts as signal when we include it in our visualisation to improve the accuracy of estimates, extract probabilities, elicit rational decisions, or encode specific meta information. In these cases, we want information about the individual distributions in and of themselves, that is, we want to extract "uncertainty" statistics such as $P(X < x)$ or $\text{Var}(X)$. Uncertainty acts as noise when we include it in our visualisation to soften false discoveries or conclusions. In these cases, we are not interested in extracting information about each individual distribution, but rather we want to see how the distribution inputs change our conclusions on the aggregate. The goal is to imbue our plots with desirable statistical properties that visually convey the statistical significance of any signal we can extract from the graphic [37].